Regulation:R23	IV/IV-B.Tech. I -Semester									
COMPUTER SCIENCE AND BUSINESS SYSTEMS										
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)										
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks	
B23CB4101	Deep Learning Techniques	PC	3	0	0	3	30	70	100	
B23HS4102	Production & Operations Management	HS	2	0	0	2	30	70	100	
#PE-IV	Professional Elective-IV	PE	3	0	0	3	30	70	100	
#PE-V	Professional Elective-V	PE	3	0	0	3	30	70	100	
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100	
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100	
B23CB4106	Prompt Engineering / SWAYAM Plus - Certificate program in Prompt Engineering and Chat GPT	SEC	0	1	2	2	30	70	100	
B23MC4101	Constitution of India	MC	2	0	0	0	30	--	30	
B23CB4107	Evaluation of Industry Internship /Mini Project.	PR	-	-	-	2	-	50	50	
		TOTAL	19	1	2	21	240	540	780	

	Course Code	Course
#PE-IV	B23CS4104	Augmented Reality & Virtual Reality
	B23CS4102	Software Architecture & Design Patterns
	B23IT4104	Blockchain Technology & Applications
	B23CB4102	Embedded Systems
	B23CB4103	Any of the12-WeekSWAYAM/NPTEL Course recommended by the BoS
#PE-V	B23CS4107	Agile Methodologies
	B23CI4109	Metaverse
	B23CS4109	Computer Vision
	B23CB4104	Data Visualization
	B23CB4105	Any of the12-WeekSWAYAM/NPTEL Course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by CE or ME or ECE or EEE or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by CE or ME or ECE or EEE or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB4101	PC	3	--	--	3	30	70	3 Hrs.

DEEP LEARNING TECHNIQUES

(for CSBS)

Course Objectives: This course is designed to:

1.	To understand the fundamentals of neural networks, perceptron models, and learning algorithms.
2.	To learn the concepts of feedforward neural networks and backpropagation for training deep models.
3.	To study advanced optimization and regularization techniques for improving neural network performance and explore convolutional neural networks for image-related applications.
4.	To understand recurrent neural networks, LSTM cells, and modern deep learning models for sequential data processing.

Course Outcomes: Upon the completion of the course, students will be able to:

S.No	Outcome	Knowledge Level
1.	**Explain** the fundamentals of neurons, perceptron models, threshold logic, and perceptron learning algorithms.	K2
2.	**Explain** feedforward networks, multilayer perceptrons, backpropagation, regularization, and autoencoders.	K2
3.	**Apply** optimization techniques, regularization methods, and CNN operations for better neural network training.	K3
4.	**Apply** RNN, LSTM, GRU, CNN architectures, and generative models for sequence and image processing tasks.	K3
5.	**Explain** recent deep learning models such as Variational Auto Encoders, Transformers, GPT, and their applications in vision, NLP, and speech.	K2

SYLLABUS

UNIT-I (10Hrs)	**Introduction to Deep Learning:** Biological Neuron, Idea of computational units, McCulloch–Pitts unit and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Linear separability, Convergence theorem for Perceptron Learning Algorithm.
UNIT-II (8 Hrs)	**Feedforward Networks-** Multilayer Perceptron, Gradient Descent, Back propagation, Empirical Risk Minimization, regularization, auto encoders. Deep Neural Networks: Difficulty of training deep neural networks, Greedy layer wise training.
UNIT-III (10 Hrs)	**Better Training of Neural Networks-** Newer optimization methods for neural networks (Adagrad, adadelta, rmsprop, adam, NAG), second order methods for training, Saddle point problem in neural networks, Regularization methods (dropout, drop connect, batch

	normalization). **Convolutional Neural Networks (CNN):** Introduction to CNNs and Their Importance, Convolution & Pooling Operations, CNN Architectures.
UNIT-IV (10 Hrs)	**Recurrent Neural Networks-** Back propagation through time, Long Short-term Memory, Gated Recurrent Units, Bidirectional LSTMs, Bidirectional RNNs. Convolutional Neural Networks: LeNet, AlexNet. Generative models: Restrictive Boltzmann Machines (RBMs), Introduction to MCMC and Gibbs Sampling, gradient computations in RBMs, Deep Boltzmann Machines.
UNIT-V (8 Hrs)	**Recent trends-** Variational Auto encoders, Transformers, GPT Applications: Vision, NLP, Speech.

Textbooks:

1.	Deep Learning, Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016.
2.	Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence, Jon Krohn, Grant Beyleveld, Aglaé Bassens, September 2019, Addison-Wesley Professional, ISBN: 9780135116821.

Reference Books:

1.	Artificial Neural Networks, Yegnanarayana, B., PHI Learning Pvt. Ltd, 2009.
2.	Deep Learning with Python, Franço is Chollet, Manning Publications,2017.
3.	Neural Networks: A Systematic Introduction, Raúl Rojas,1996.

e-Resources

1.	Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc22_cs22/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4102	HS	2	--	--	2	30	70	3 Hrs.

PRODUCTION & OPERATIONS MANAGEMENT

(For CIC & CSBS)

Course Objectives:

1.	To impart a comprehensive understanding of Production &Operations Management principles and practices.
2.	To equip students with knowledge of product and process design principles, innovation, development, and optimization techniques.
3.	To provide knowledge of supply chain strategies, inventory management, logistics, demand forecasting, and operational efficiency
4.	To develop understanding of quality management principles, continuous improvement methods, process optimization, and performance enhancement strategies.
5.	To equip students with knowledge of operations planning, scheduling, control techniques, resource management, and process optimization.

Course Outcomes: At the end of the Course, students will be able to

S.No	Outcome	Knowledge Level
1.	**Apply** the principles of Production &**Operations Management** in analyzing production processes and operational activities in organizations	K3
2.	**Apply** the principles of product and process design, innovation, optimization, sustainability, and manufacturing efficiency.	K3
3.	**Implement** qualitative and quantitative demand forecasting techniques for supply chain planning.	K3
4.	**Illustrate** the importance of global quality standards such as ISO series for organizational compliance.	K3
5.	**Implement** strategic production planning and operational scheduling to improve organizational productivity.	K3

SYLLABUS

UNIT-I (8 Hrs)	**Introduction to Production &Operations Management:** Definition, Nature, and Scope of Operations, Variables influencing the Production & Operations Management Management : Role and Significance; Evolution of Operations Management: Historical Perspectives; Operations Strategy: Formulation and Implementation
UNIT-II (8 Hrs)	**Product and Process Design:** Product Development Life Cycle and Innovation Strategies; Production System Classifications: Job, Batch, Mass, and Continuous Production; Facility Design: Floor Plan Mapping & types; Location Optimization & Site Selection Criteria
UNIT-III (12 Hrs)	**Supply Chain and Inventory Management:** Overview of Supply Chain Management: Concepts and Importance; Inventory Control Systems: Economic Order Quantity, ABC

	Analysis, Just In Time; Demand Forecasting Techniques: Qualitative and Quantitative Models; Digital Supply Chains: Role of Technology and Automation
UNIT-IV (10Hrs)	**Quality Management and Improvement:** Fundamentals of Quality Management and TQM Principles, Quality Assurance Tools: Six Sigma, Kaizen, SPC, and Benchmarking; Global Quality Standards: ISO Series and Compliance.
UNIT-V (10Hrs)	**Operations Planning and Control**: Strategic Production Planning and Operational Scheduling; Capacity Planning Techniques and Resource Optimization; Project Management Methodologies: Critical Path Method; Program Evaluation and Review Technique

Textbooks:	
1.	Aswathappa, K., & Bhattacharya, S. (2010). Production and Operations Management. Himalaya Publishing House
2.	Panneerselvam, R. (2012). Production and Operations Management. PHI Learning
3.	KanishkaBedi (2018). Production and Operations Management. Viva Books.
Reference Books:	
1.	Heizer, J., Render, B., & Munson, C. (2017). Operations Management. Pearson Education.
2.	Stevenson, W. J. (2020). Operations Management. McGraw-Hill Education.
3.	Chase, R. B., Jacobs, F. R., &Aquilano, N. J. (2019). Operations and Supply Chain Management. McGraw-Hill.
4.	Slack, N., & Brandon-Jones, A. (2019). Operations Management. Pearson.
5.	Krajewski, L. J., Malhotra, M. K., & Ritzman, L. P. (2018). Operations Management: Processes and Supply Chains. Pearson Education.
e-Resources	
1.	https://nptel.ac.in/courses/112107238

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4104	**PE**	**3**	**--**	**--**	3	**30**	**70**	**3 Hrs.**

AUGUMENTED REALITY & VIRTUAL REALITY

(Common to CSE & CSBS)

Pre-requisites: Image Processing, Computer Vision and Machine Vision

Course Objectives: Students are expected

1.	To define augmented reality (AR) and its related fields, which include virtual reality (VR) and mixed reality (MR). Analyze the display requirements for AR and the technology used for tracking. Apply AR concepts to solve real-world problems, and evaluate the usability and effectiveness of AR solutions.
2.	To provide students with knowledge and skills for developing computer vision techniques and interaction methods specifically for Augmented Reality (AR) applications.
3.	To familiarize students with the terminology, background, human perception, and technical underpinnings of virtual reality (VR).
4.	To comprehend the physiology of human vision and how it affects virtual reality (VR), including rendering methods, visual perception, and VR experience optimization.
5.	To understand interaction strategies, audio rendering for immersive experiences, and motion perception in both real and virtual environments.

Course OutComes: Upon completing this course, student will learn:

S. No	**OUTCOME**	KL
1.	Analyze differences and similarities in augmented reality systems that assess environments, tackle industry challenges, and evaluate serviceability, effectiveness, user experience, latency, field of view, and display types.	**K3**
2.	Use markers to position AR objects. Use infrared cameras to track them. Handle objects without markers and solve outdoor AR problems. Use voice and gestures for input. Use visuals and sound for output. Focus on combining AR with real objects and meeting AR app design needs.	**K3**
3.	Identify VR applications, connect VR design to human sensations, depict 3D objects and changes, and utilize light, lens, and display technologies involving cameras and vision in virtual reality.	**K3**
4.	Understand visual pathways from cornea to cortex, implications for VR design, depth, motion, and color perception in VR, as well as rasterization, shading, ray tracing, latency, frame rates, and distortions.	**K3**
5.	Analyzing motion perception, managing VR physics, addressing motion mismatch, designing interaction techniques, and applying audio rendering principles.	**K3**

SYLLABUS

UNIT-I (10 Hrs)	Introduction to Augmented Reality: Augmented Reality - Defining augmented reality, history of augmented reality, Examples, Related fields. Displays: Multimodal Displays, Visual Perception, Requirements and Characteristics, Spatial Display Model, Visual

	Displays. Tracking: Tracking, Calibration, and Registration, Coordinate Systems, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors.
UNIT-II (10 Hrs)	Computer Vision for Augmented Reality: Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Outdoor Tracking. Interaction: Output Modalities, Input Modalities, Tangible Interfaces, Virtual User Interfaces on Real Surfaces, Augmented Paper, Multi-view Interfaces, Haptic Interaction. Software Architectures: AR Application Requirements, Software Engineering Requirements, Distributed Object Systems, Dataflow, Scene Graphs.
UNIT-III (10 Hrs)	Introduction to Virtual Reality: Defining Virtual Reality, History of VR, Human Physiology and Perception. The Geometry of Virtual Worlds: Geometric Models, Axis-Angle Representations of Rotation, Viewing Transformations. Light and Optics: Basic Behavior of Light, Lenses, Optical Aberrations, The Human Eye, Cameras, Displays.
UNIT-IV (10 Hrs)	The Physiology of Human Vision: From the Cornea to Photoreceptors, From Photoreceptors to the Visual Cortex, Eye Movements, Implications for VR. Visual Perception: Visual Perception - Perception of Depth, Perception of Motion, Perception of Colo Visual Rendering: Visual Rendering -Ray Tracing and Shading Models, Rasterization, Correcting Optical Distortions, Improving Latency and Frame Rates, Immersive Photos and Videos.
UNIT-V (10 Hrs)	Motion in Real and Virtual Worlds: Velocities and Accelerations, The Vestibular System, Physics in the Virtual World, Mismatched Motion and Vection. Interaction: Motor Programs and Remapping, Locomotion, Social Interaction. Audio: The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering.

TEXTBOOK:

1.	**"Augmented Reality: Principles & Practice" by Schmalstieg, Hollerer, Pearson Education India; First edition (12 October 2016), ISBN-10: 9332578494.**
2.	**Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016.**

REFERENCE BOOKS:

1.	"AR Game Development", Allan Fowler, 1st Edition, Apress Publications, 2018, ISBN 978-1484236178.
2.	"Understanding Virtual Reality: Interface, Application and Design", William R Sherman and Alan B Craig, (The Morgan Kaufmann Series in Computer Graphics)". Morgan Kaufmann Publishers, San Francisco, CA, 2002.
3.	"Developing Virtual Reality Applications: Foundations of Effective Design", Alan B Craig, William R Sherman and Jeffrey D Will, Morgan Kaufmann, 2009.
4.	"Designing for Mixed Reality", Kharis O'Connell, O'Reilly Media, Inc., 2016, ISBN:9781491962381.
5.	"Theory and applications of marker-based augmented reality", Sanni Siltanen, Julkaisija, Utgivare Publisher. 2012. ISBN 978-951-38-7449-0.
6.	"Designing Virtual Systems: The Structured Approach", Gerard Jounghyun Kim, 2005.

E-Resources:	
1.	https://thearea.org/augmented-reality/
2.	https://www.vntana.com/blog/creating-augmented-reality-content/
3.	https://wayup.in/en/blog/augmented-reality-ar-source-of-inspiration-and-ideas-for-web-desig

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4102	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

SOFTWARE ARCHITECTURE & DESIGN PATTERNS

(Common to CSE, AIDS, CSBS, CSD)

Pre-requisites: OOP,SE,OOAD

Course Objectives: Students are expected

1.	Understand the basic concepts to identify state behaviour of real world objects
2.	Apply Object Oriented Analysis and Design concepts to solve complex problems
3.	Construct various UML models using the appropriate notation for specific problem context
4.	Design models to Show the importance of systems analysis and design in solving complex problems using case studies
5.	Study Pattern Oriented approach for real world problems

Course OutComes: At the end of the course students will be able to

S. No	outcome	KL
1.	**Explain** the concepts of design patterns and object-oriented development.	2
2.	**Use** object-oriented analysis techniques to gather requirements and identify conceptual classes and relationships in a system.	3
3.	**Demonstrate** structural design patterns such as Adapter, Bridge, Composite, Decorator, Facade, Flyweight, and Proxy.	3
4.	**Illustrate** the MVC architectural pattern in designing interactive systems.	3
5.	**Apply** distributed object system concepts including client–server architecture, Java RMI, and web services.	3

SYLLABUS

UNIT-I (10 Hrs)	Introduction: design pattern, Describing design patterns, the catalog of design pattern, organizing the catalog, how design patterns solve design problems, how to select a design pattern, how to use a design pattern. What is object oriented development-key concepts of object oriented design other related concepts, benefits and drawbacks of the paradigm
UNIT-II (10 Hrs)	Analysis a System: Overview of the analysis phase, stage 1 gathering the requirements functional requirements specification, defining conceptual classes and relationships, using the knowledge of the domain Design and Implementation, discussions and further reading
UNIT-III (10 Hrs)	Design Pattern Catalog: Structural patterns, Adapter, bridge, composite, decorator, facade, Flyweight, proxy.
UNIT-IV (10 Hrs)	Interactive systems and the MVC architecture: Introduction The MVC architectural pattern, analyzing a simple drawing program designing the system, designing of the subsystems, getting into implementation, implementing undo operation drawing

	incomplete items, adding a new feature pattern based solutions
UNIT-V (10 Hrs)	Designing with Distributed Objects: Client server system, java remote method invocation, implementing an object oriented system on the web, Web services (SOAP, Restful), Enterprise Service Bus
Text books:	
1.	Object Oriented Analysis, Design and Implementation, Brahma Dathan, Sarnath Rammath , Universities Press, 2013
2.	Design Patterns, Erich Gamma, Richard Helan, Ralph Johman, John Vlissides, PEARSON Publication, 2013
Reference books:	
1.	Frank Bachmann, Regine Meunier, Hans Rohnert "Pattern Oriented Software Architecture", Volume 1, 1996.
2.	William J Brown et al., "Anti Patterns: Refactoring Software, Architectures and Projects in Crisis", John Wiley, 1998
E-Resources:	
1.	Object Oriented Analysis and Design By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay, Prof. Kausik Datta | IIT Kharagpur https://onlinecourses.nptel.ac.in/noc21_cs57/preview
2.	Object Oriented System Development Using UML, Java And Patterns, IIT Kharagpur Prof. Rajib Mall https://nptel.ac.in/courses/106105224

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4104	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

BLOCKCHAIN TECHNOLOGY & APPLICATIONS

(Common to IT,AIDS, CSBS)

Course Objectives: This course is designed to:

1.	Understand the fundamental concepts of blockchain technology, its architecture, and working principles.
2.	Explain different types of blockchain systems and consensus mechanisms used in distributed environments.
3.	Analyze public, private, and consortium blockchain platforms and their real-world implementations.
4.	Explain and analyze cryptocurrency systems and smart contract mechanisms.
5.	Analyze security, privacy challenges, and performance issues in blockchain technology.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Explain** blockchain fundamentals, architecture, and cryptographic concepts	K2
2.	**Explain** smart contracts and cryptocurrency transaction mechanisms in blockchain platforms	K3
3.	**Analyze** private and consortium blockchain systems and permissioned blockchain algorithms	K2
4.	**Analyze** security, privacy challenges and applications of blockchain in emerging real-world scenarios.	K3
5.	**Analyze** blockchain case studies and implement blockchain applications using python and hyperledger fabric.	K3

SYLLABUS

UNIT-I	**Fundamentals of Blockchain:** Introduction, Origin of Blockchain, Blockchain Solutions, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency: Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.
UNIT-II	**Public Blockchain System:** Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.

UNIT-III	**Private Blockchain System:** Introduction, Key Characteristics of Private Blockchain, Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain. Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Need of Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda.
UNIT-IV	**Security in Blockchain**: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric. **Applications of Blockchain:** Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.
UNIT-V	**Blockchain Case Studies:** Case Study 1 – Retail, Case Study 2 – Banking and Financial Services, Case Study 3 – Healthcare, Case Study 4 – Energy and Utilities. Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK.

Textbooks:	
1.	"Blockchain Technology", Chandramouli Subramanian, Asha A.George, Abhilasj K A, Meena Karthikeyan , Universities Press.
2.	Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gauar, Pearson Addition Wesley
Reference Books:	
1.	Blockchain Blue print for Economy, Melanie Swan, SPD Oreilly.
e-Resources:	
1	NPTEL: Blockchain Architecture Design and Use Cases – https://nptel.ac.in
2	Ethereum Documentation – https://ethereum.org
3	Hyperledger Documentation – https://hyperledger.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB4102	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

EMBEDDED SYSTEMS

(for CSBS)

Course Objectives: This course is designed to:

1.	To understand the fundamentals of embedded systems and processor selection for embedded applications.
2.	To learn embedded firmware design methodologies and development tools.
3.	To study Real-Time Operating Systems (RTOS) concepts including task scheduling, communication, and synchronization.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Explain** the fundamentals, characteristics, classifications, and application areas of embedded systems.	**K2**
2.	**Describe** the architecture of embedded systems including processors, memory, sensors, actuators, and communication interfaces.	**K2**
3.	**Apply** embedded firmware design approaches and development languages for basic embedded system design.	**K3**
4.	**Explain** mechanisms for efficient task communication and synchronization, including shared memory, message passing, RPC, and sockets.	**K2**
5.	**Explain** device drivers and key criteria for selecting an appropriate RTOS based on system requirements.	**K2**

SYLLABUS

UNIT-I (10Hrs)	**Introduction to Embedded Systems:** Definition of Embedded System, Embedded Systems Vs General Computing Systems, History of Embedded Systems, Classification, Major Application Areas, Purpose of Embedded Systems, Characteristics and Quality Attributes of Embedded Systems.
UNIT-II (8 Hrs)	**Typical Embedded System:** Core of the Embedded System, General Purpose and Domain Specific Processors, ASICs, PLDs, Commercial Off-The-Shelf Components (COTS), memory: ROM, RAM, Memory according to the type of Interface, Memory Shadowing, Memory selection for Embedded Systems, Sensors and Actuators, Communication Interface: Onboard and External Communication Interfaces.
UNIT-III (10 Hrs)	**Embedded Firmware:** Reset Circuit, Brown-out Protection Circuit, Oscillator Unit, Real Time Clock, Watchdog Timer, Embedded Firmware Design Approaches and Development Languages.

UNIT-IV (10 Hrs)	**RTOS Based Embedded System Design:** Operating System Basics, Types of Operating Systems, Tasks, Process and Threads, Multiprocessing and Multitasking, Task Scheduling.
UNIT-V (8 Hrs)	**Task Communication:** Shared Memory, Message Passing, Remote Procedure Call and Sockets, **Task Synchronization:** Task Communication/Synchronization Issues, Task Synchronization Techniques, Device Drivers, Methods to Choose an RTOS.
Textbooks:	
1.	Introduction to Embedded Systems - Shibu K.V, Mc Graw Hill.
2.	Embedded Systems - Raj Kamal, TMH.
Reference Books:	
1.	Embedded System Design - Frank Vahid, Tony Givargis, John Wiley.
2.	Embedded Systems – Lyla, Pearson, 2013

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4107	**PE**	**3**	**--**	**--**	3	**30**	**70**	**3 Hrs.**

AGILE METHODOLOGIES

(For CSE, AIML, AIDS, CSBS, IT)

Pre-requisites: Software Engineering

Course Objectives: Students are expected

1.	Gain deep comprehension of Agile philosophy, principles, and values to support high-quality software delivery.
2.	Employ different Agile frameworks (Scrum, XP, Lean, Kanban) to real-world projects for improved adaptability, productivity, and customer satisfaction.

Course Out Comes: At the end of the course students will be able to

S. No	OUTCOME	KL
1.	Apply agile values and principles to foster team collaboration and adaptability in projects.	K3
2.	Demonstrate the 12 agile principles to direct project delivery in changing contexts.	K3
3.	Use Scrum roles, events, and artifacts to oversee iterative development cycles.	K3
4.	Model XP practices to boost software quality and team responsiveness.	K3
5.	Determine Lean and Kanban principles to detect waste and streamline development workflows.	K3

SYLLABUS

UNIT-I (10 Hrs)	**Learning Agile:** Agile, Getting Agile into your brain, **Understanding Agile values**: No Silver Bullet, Agile to the Rescue. A fractured perspective, The Agile Manifesto, Understanding the Elephant, Where to Start with a New Methodology.
UNIT-II (10 Hrs)	**The Agile Principles:** The 12 Principles of Agile Software, The Customer Is Always Right, Delivering the Project, Better Project Delivery for the Ebook Reader Project. Communicating and Working Together, Project Execution—Moving the Project Along, Constantly Improving the Project and the Team. The Agile Project: Bringing All the Principles Together
UNIT-III (10 Hrs)	**SCRUM and Self-Organizing Teams:** The Rules of Scrum, Act I: I Can Haz Scrum, Everyone on a Scrum Team owns the Project, Status Updates Are for Social Networks!, The Whole Team Uses the Daily Scrum, Feedback and the Visibility-Inspection-Adaptation Cycle, The Last Responsible Moment, Sprinting into a Wall, Sprints, Planning, and Retrospectives. **Scrum Planning And Collective Commitment:** Not Quite Expecting the Unexpected, User

	Stories, Velocity, and Generally Accepted Scrum Practices, Victory Lap, Scrum Values Revisited.
UNIT-IV (10 Hrs)	**XP And Embracing Change:** Going into Overtime, The Primary Practices of XP, The Game Plan Changed, but We're Still Losing, The XP Values Help the Team Change Their Mindset, An Effective Mindset Starts with the XP Values, The Momentum Shifts, Understanding the XP Principles Helps You Embrace Change. **XP, Simplicity, and Incremental Design:** Code and Design, Make Code and Design Decisions at the Last Responsible Moment, Final Score.
UNIT-V (10 Hrs)	**Lean, Eliminating Waste, and Seeing the whole:** Lean Thinking, Creating Heroes and Magical Thinking. Eliminate Waste, Gain a Deeper Understanding of the Product, Deliver As Fast As Possible. **Kanban, Flow, and Constantly Improving:** The Principles of Kanban, Improving Your Process with Kanban, Measure and Manage Flow, Emergent Behavior with Kanban. **The Agile Coach:** Coaches Understand Why People Don't Always Want to Change. The Principles of Coaching.
Textbook:	
1.	**Andrew Stellman, Jennifer Greene, Learning Agile, O'Reilly, 2015.**
Reference books:	
1.	Andrew Stellman, Jennifer Green, Head first Agile, O'Reilly, 2017.
2.	Rubin K , Essential Scrum : A Practical Guide To The Most Popular Agile Process, Addison-Wesley, 2013
E-Resources:	
1.	https://www.agilealliance.org
2.	https://www.scrum.org
3.	https://www.scrumtrainingseries.com
4.	https://www.agilemanifesto.org
5.	https://www.scrumguides.org

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CI4109	**PE**	**3**	**--**	**--**	**3**	**30**	**70**	**3 Hrs.**

METAVERSE

(Common to CIC & CSBS)

Pre-requisites: Basic computer knowledge, Internet, Networking and Programming basics

Course Objectives: Students are expected

1.	To understand the fundamental concepts of the Metaverse, including its architecture, technologies, and applications.
2.	To explore Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) as core components of the Metaverse.
3.	To study blockchain, NFTs, and decentralized finance (DeFi) as economic foundations of the Metaverse.
4.	To analyze Metaverse-based social interactions, business models, and ethical considerations
5.	To gain hands-on experience with tools and platforms used to develop Metaverse applications.

Course Out Comes: At the end of the course students will be able to

S. No	OUT COME	KL
1.	Apply the concepts of Metaverse architecture and enabling technologies to develop simple use cases or prototypes for real-world applications such as education, gaming, or virtual collaboration.	K3
2.	Apply the concepts of Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) to develop and demonstrate immersive applications as core components of the Metaverse.	K3
3.	Apply the concepts of Blockchain, NFTs, and Decentralized Finance (DeFi) to design and demonstrate digital economic models within the Metaverse.	K3
4.	Analyse Metaverse-based social interactions, business models, and ethical considerations to understand their impact on users and digital ecosystems.	K4
5.	Apply development tools and platforms to build and demonstrate basic Metaverse applications, gaining hands-on experience in immersive environment creation.	K3

SYLLABUS

UNIT-I (10 Hrs)	Introduction to Metaverse: Definition and Evolution of the Metaverse, Key Components: VR, AR, MR, AI, and Blockchain, Applications of Metaverse: Gaming, Education, Healthcare, and Business. Challenges and Ethical Issues in the Metaverse
UNIT-II (10 Hrs)	Virtual Reality (VR) and Augmented Reality (AR): Fundamentals of VR and AR Technologies, Hardware and Software Requirements Interaction Techniques and User Experience, Metaverse Platforms: Oculus, Microsoft Mesh, Horizon Worlds, Hands-on: Creating a Simple VR/AR Environment
UNIT-III (10 Hrs)	Blockchain and Decentralization in the Metaverse: Introduction to Blockchain Technology, Cryptocurrencies, NFTs, and Smart Contracts, Decentralized Applications (DApps) and

	Web3, Security and Privacy considerations Hands-on: Deploying an NFT on a Test Blockchain
UNIT-IV (10 Hrs)	Metaverse Economy and Social Structures: Digital Goods, Virtual Real Estate, and Digital Identity, Economic Models: Play- to-Earn, DeFi, and Token omics, Social and Ethical Aspects: Digital Citizenship and Governance, Future Trends in Metaverse Economy
UNIT-V (10 Hrs)	Tools and Development in the Metaverse: Metaverse Development Platforms: Unity, Unreal Engine, and WebXR, AI and Cloud Computing in the Metaverse, 3D Asset Creation and Interoperability Standards, Hands-on: Building a Basic Metaverse Application
Text books:	
1.	Pallavi Agarwal, Metaverse Fundamentals: Easy Hands-on Book on Understanding Metaver Buying Land, NFTs, Virtual Reality, Augmented Reality, Blockchain & Crypto Art. Notion pre Publisher.
2.	Tony Parisi, Learning Virtual Reality: Developing Immersive Experiences and Applications for desktop, Web, and Mobile, O'Reilly Media.
Reference Books:	
1.	Matthew Ball, The Metaverse: And How It Will Revolutionize Everything, Liveright Publishing.
2.	Lalit Kumar Awasthi, Sanjay Kumar, Metaverse: Concept, Content and Context , Springer 2023
3.	William R. Sherman, Understanding Virtual Reality, Morgan kaufmann-2nd edition 2018
4.	Daniel Drescher, Blockchain Basics, Apress -2017
5.	Mark Van Rijmenam, Step into the Metaverse: How the Immersive Internet Will Unlock a Trillion-Dollar Social Economy, Wiley.
6.	Josh O'Kane, The Metaverse Economy: How Web3, Blockchain, and AI Are Shaping the Future of Business.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4109	**PE**	**3**	**--**	**--**	3	**30**	**70**	**3 Hrs.**

COMPUTER VISION

(Common to CSE, CSBS)

Pre-requisites: Solid foundation in Python programming, linear algebra, calculus, and basics of machine learning/deep learning

Course Objectives: Students are expected

1.	Understand the fundamentals of camera models and image formation.
2.	Develop knowledge of image processing techniques.
3.	Understand the geometry of multiple views and stereopsis.
4.	Learn segmentation and model fitting techniques.
5.	Learn geometric camera models and calibration techniques and Understand model-based vision approaches.

Course Out Comes: At the end of the course students will be able to

S. No	OUTCOMES	Knowledge Level
1.	**Understand** the **principles of image formation and camera models**, including pinhole cameras, radiometry, shading models, and color representation in computer vision.	K2
2.	**Apply image processing techniques** such as linear filtering, convolution, Fourier transforms, edge detection, and texture analysis for image understanding.	K3
3.	**Analyze multiple view geometry and stereopsis** to perform depth estimation and apply segmentation techniques using clustering methods.	K4
4.	**Implement segmentation and model fitting techniques** including Hough Transform, probabilistic inference methods, EM algorithm, and tracking using linear dynamic models and Kalman filtering.	K3
5.	**Apply geometric camera calibration and model-based vision techniques** for applications such as robot localization, medical image registration, and object recognition.	K3

SYLLABUS

UNIT-I (10 Hrs)	Cameras: Pinhole Cameras Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Interreflections: Global Shading Models Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.
UNIT-II	Linear Filters: Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial

(10 Hrs)	Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates, Edge Detection: Noise, Estimating Derivatives, Detecting Edges Texture0: Representing Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.
UNIT-III (10 Hrs)	The Geometry of Multiple Views: Two Views Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and Getstalt, Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph Theoretic Clustering,
UNIT-IV (10 Hrs)	Segmentation by Fitting a Model: The Hough Transform, Fitting Lines, Fitting Curves, fitting as a Probabilistic Inference Problem, Robustness Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice, Tracking with Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples
UNIT-V (10 Hrs)	Geometric Camera Models: Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, Case study: Mobile Robot Localization Model- Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtaining Hypotheses by pose Clustering, Obtaining Hypotheses Using Invariants, Verification, Case study: Registration in Medical Imaging Systems, Curved Surfaces and Alignment.

Textbook:

1.	**David A. Forsyth, Jean Ponce, "Computer Vision – A Modern Approach", PHI Learni (Indian Edition), 2009.**
2.	**E. R. Davies, "Computer and Machine Vision – Theory, Algorithms and Practicalitie Elsevier (Academic Press), 4th edition, 2013.**

Reference books:

1.	R. C. Gonzalez, R. E. Woods, "Digital Image Processing", Addison Wesley, 2008.
2.	Richard Szeliski "Computer Vision: Algorithms and Applications" Springer, Verlag London Limited, 2011.

E-Resource:

1.	https://onlinecourses.nptel.ac.in/noc26_ee74/preview
2.	https://onlinecourses.nptel.ac.in/noc26_ee31/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB4104	**PE**	**--**	**1**	**2**	**2**	**30**	**70**	**3 Hrs.**

DATA VISUALIZATION

(for CSBS)

Course Objectives: This course is designed to:

1.	Familiarize students with the basic and advanced techniques information visualization and scientific visualization
2.	Learn key techniques of the visualization process
3.	A detailed view of visual perception, the visualized data and the actual visualization, interaction and distorting techniques

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Explain** Visualization and representation of data	**K2**
2.	**Creating** visual representations and visualization reference model of applications	**K3**
3.	**Classify** the visualization systems in a data representation	**K2**
4.	**Identify** Visualization of groups and trees	**K3**
5.	**Determine** the visualization of volumetric different data sets in applications	**K2**

SYLLABUS

UNIT-I (10Hrs)	**Introduction**: What Is Visualization? History of Visualization, Relationship between visualization and Other Fields. The Visualization Process, Introduction of visual perception, visual representation of data, Gestalt principles and information overloads
UNIT-II (8 Hrs)	Creating visual representations, visualization reference model, visual mapping, visual analytics, Design of visualization applications.
UNIT-III (10 Hrs)	Classification of visualization systems, Interaction and visualization techniques misleading, Visualization of one, two and multi-dimensional data, text and text documents.
UNIT-IV (10 Hrs)	Use of visualization software to create charts and graphical representations. Visualization of groups, trees, graphs, clusters, networks, software, Metaphorical visualization
UNIT-V (8 Hrs)	Visualization of volumetric data, vector fields, processes and simulations, Visualization of maps, geographic information, GIS systems, collaborative visualizations, evaluating visualizations. Recent trends in various perception techniques, various visualization techniques, data structures used in data visualization

Textbooks:

1.	WARD, GRINSTEIN, KEIM. Interactive Data Visualization: Foundations, Techniques, and Applications. Natick : A K Peters, Ltd.
2.	E. Tufte, The Visual Display of Quantitative Information, Graphics Press.
Reference Books:	
1.	https://kdd.cs.ksu.edu/Courses/CIS536/Lectures/Slides/Lecture-34-Main_6up.pdf

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB4106	**SEC**	**--**	**1**	**2**	**2**	**30**	**100**	**3 Hrs**

PROMPT ENGINEERING

(for CSBS)

Course Objectives: This course is designed to:

1.	Introduce the fundamental concepts of prompt engineering, including proper design principle, interactive refinement and interaction with LLMs
2.	Ability to design and apply advanced prompting techniques
3.	enable students to build , evaluate, and deploy prompt driven applications

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Apply** iterative prompting for clarity and context.	**K4**
2.	**Create** varied prompts to steer model outputs.	**K4**
3.	**Construct** chain-of-thought and structured prompts.	**K3**
4.	**Develop** retrieval-augmented pipelines to ground outputs.	**K4**
5.	**Evaluate** LLM agents and multimodal apps for ethics and robustness.	**K4**

SYLLABUS

UNIT-I (10Hrs)	**Foundations of Prompt Engineering:** Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design, Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation. **Lab Experiments:** 1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple “Hello, world” prompt to verify model access. 2. Baseline vs. Enhanced Prompts: Execute a naïve prompt (“Write a one-paragraph bio of Ada Lovelace.”) and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style.

UNIT-I (10Hrs)	3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean play Romeo and Juliet in two sentences through three rounds of prompt tweaking: a. Minimal instruction. b. Addition of length and style constraints c. Specification of key content elements (setting and theme) Document how each iteration changes and improves the result. 4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt; analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions.
UNIT-II (8 Hrs)	**Advanced Prompt Patterns & Techniques:** Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization. **Lab Experiments:** 1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples. 2. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., “You are a financial advisor...”); then create a negative prompt to suppress undesired content (e.g., “Do not mention any brand names”); evaluate how each influences the model’s response. 3. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.

UNIT-III (10 Hrs)	**Structured Output & Reasoning Techniques:** Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks **Lab Experiments:** 1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., "List three benefits of daily exercise in a Markdown table with columns 'Benefit' and 'Description.'"); verify the output matches the requested structure. 2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. 3. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic puzzle) and apply zero-shot CoT prompting (e.g., "Let's think step by step. Explain your reasoning before the final answer."); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline.
UNIT-IV (10 Hrs)	**Retrieval-Augmented Generation & LangChain Workflows:** Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval- Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval- generation pipeline with an LCEL chain **Lab Experiments:** 1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., "Summarize this text: ..."), passes it to an LLM, and prints the result; verify end-to-end execution. 2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in-memory vector store; inspect for consistency. 3. Constructing & Running a Basic RAG Chain: Build a pipeline that: a. Receives a user query b. Retrieves the top-k relevant chunks c. Constructs a combined prompt with context + query d. Send it to the LLM e. Returns the answer e. Test with sample queries and compare factual accuracy against a prompt without retrieval.

UNIT-V (8 Hrs)	**Agents, Multimodal AI & Ethical Evaluation:** Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to "LLM-as-Judge" for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning **Lab Experiments:** 1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. 2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt to evaluate alignment. 3. Prompt Evaluation & Ethics Workshop: a. Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. b. Use an "LLM-as-Judge" prompt (e.g., "Rate these outputs on a scale of 1–5 for clarity and correctness.") to automate evaluation. c. Design a prompt- injection test (e.g., "Ignore previous instructions..."), observe the response, then refine system prompts to mitigate the vulnerability.

Textbooks:

1.	James Phoenix, Mike Taylor, "Prompt Engineering for Generative AI", O'Reilly, To Release in May 2024 https://www.oreilly.com/library/view/prompt-engineering-for/9781098153427/
2.	Gilbert Mizrahi, "Unlocking the Secrets of Prompt Engineering: Master the Art of Creative Language Generation to Accelerate Your Journey from Novice to Pro", January 2024 https://www.packtpub.com/en-in/product/unlocking-the-secrets-of-prompt-engineering9781835083833

Reference Books:

1.	Michael Ferguson, "Prompt Engineering: The Future of Language Generation", January 2023 https://books.apple.com/us/book/prompt-engineering-the-future-of-languagegeneration/id6445529200
2.	"Prompt Engineering Guide", https://www.promptingguide.ai/
3.	"Prompt Engineering for Generative AI", Google, https://developers.google.com/machinelearning/resources/prompt-eng

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	-	3 Hrs.

CONSTITUTION OF INDIA

(for CSBS)

Course Objectives: This course is designed to:

1.	Understand the premises informing the twin themes of liberty and freedom from a civil rights perspective.
2.	To address the growth of Indian opinion regarding modern Indian intellectuals' constitutional role and entitlement to civil and economic rights as well as the emergence of nationhood in the early years of Indian nationalism.
3.	To address the role of socialism in India after the commencement of the Bolshevik Revolution in 1917 and its impact on the initial drafting of the Indian Constitution.

Course Outcomes: Upon the completion of the course students will be able to:

S.No	Outcome	Knowledge Level
1.	**Understand** the historical evolution, drafting process, and philosophical foundations (Preamble and salient features) of the Indian Constitution.	**K2**
2.	**Apply** knowledge of Fundamental Rights, Directive Principles of State Policy, and Fundamental Duties to analyze contours of constitutional rights and duties.	**K3**
3.	**Apply** understanding of the structure, powers, and functions of Parliament, Executive, and Judiciary in India's governance framework.	**K3**
4.	**Understand** the organization, roles, and significance of local administration institutions like Panchayati Raj, municipalities, and district administration.	**K2**
5.	**Apply** concepts of the Election Commission's functioning and institutions for SC/ST/OBC/women welfare to evaluate democratic processes and inclusivity.	**K3**

SYLLABUS

UNIT-I (10Hrs)	**History of Making of the Indian Constitution:** History, Drafting Committee, (Composition & Working) Philosophy of the Indian Constitution- Preamble, Salient, Features
UNIT-II (8 Hrs)	**Contours of Constitutional Rights & Duties:** Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties.
UNIT-III (10 Hrs)	**Organs of Governance:** Parliament, Composition, Qualifications and Disqualifications, Powers and Functions, **Executive-** President, Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions

UNIT-IV **(10 Hrs)**	**Local Administration:** District's Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative CEO of Municipal Corporation, Pachayati raj: Introduction, PRI: ZilaPachayat, Elected officials and their roles, CEO ZilaPachayat: Position and role, Block level: Organizational Hierarchy (Different departments), Village level: Role of Elected and Appointed officials, Importance of grass root democracy
UNIT-V **(8 Hrs)**	**Election Commission:** Role and Functioning, Chief Election Commissioner and Election Commissioners, **State Election Commission**: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	The Constitution of India, 1st Edition, (Bare Act), Government Publication, 1950
2.	Framing of Indian Constitution, 1st Edition, Dr. S. N. Busi, Dr. B. R. Ambedkar 2015
Reference Books:	
1.	Indian Constitution Law, 7th Edition, M. P. Jain, Lexis Nexis, 2014

Regulation:R23	IV/IV-B.Tech.II -Semester								
COMPUTER SCIENCE AND BUSINESS SYSTEMS									
COURSE STRUCTURE (Witheffectfrom2023-24admittedBatchonwards)									
Course Code	**Course Name**	**Category**	**L**	**T**	**P**	**Cr**	**C.I.E.**	**S.E.E.**	**Total Marks**
B23CB4201	Project and Internship	**PR**	**-**	**-**	**24**	**12**	**60**	**140**	**200**

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CB4201	PR	--	--	24	12	60	140	3 Hrs.

PROJECT WORK
(For CSBS)

Course Objectives: This course is designed to:

1	To enable students to independently research, analyze, and formulate a well-defined engineering problem by reviewing contemporary literature and technical databases.
2	To guide students in conceptualizing, designing, and optimizing innovative solutions, components, or processes while balancing realistic socio-economic and technical constraints.
3	To train students to build, test, and troubleshoot physical prototypes or software systems through the self-taught application of modern engineering tools and simulation software.
4.	To encourage students to evaluate the environmental, safety, legal, and cultural impacts of their engineering solutions while maintaining professional ethics.
5.	To nurture collaborative teamwork, project resource management, technical report writing, and effective multi-media presentation skills.

Course Outcomes: Upon the completion of the course students will be able to:

	Outcome	Knowledge Level
1.	**Identify and formulate** a complex engineering problem by reviewing literature and conducting feasibility studies.	**K4**
2.	**Design and develop** an innovative, safe, and sustainable solution, component, or process using modern engineering tools.	**K5**
3.	**Implement and test** the designed solution, analyzing the resulting data to draw valid engineering conclusions.	**K4**
4.	**Evaluate** the impact of the project solution on society, environment, and health, incorporating ethical considerations.	**K5**
5.	**Communicate** effectively through technical reports, presentations, and project demonstrations.	**K3**
6	**Demonstrate** teamwork, leadership, and project management skills, adhering to ethical practices and financial constraints.	**K3**

*The object of Project Work is to enable the student to take up investigative study in the broad field of, Computer Science And Business Systems either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work.

The assignment to normally include:

Survey and study of published literature on the assigned topic.